MOHAMMAD ZAID MOONSAMY

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RESEARCH INTERESTS

I work at the intersection of artificial intelligence (AI) and global health. My research focuses on training, optimising and applying AI agents and language models to problems in healthcare and the biomedical domain, from AI-assisted scientific discovery to offline clinical decision support in low-resource settings.

EDUCATION

Master of Science in Computer Science (by Dissertation) **Jan 2025 - Present**
University of the Witwatersrand

- Thesis (working title): "From Relevance to Reliability: Small Language Models, Contradiction Detection and Large-Scale Deployment for Microbe–Cancer Pair Discovery". Supervised by Prof Bruce Bassett, Prof Richard Klein and Dr Hairong Wang.
- Member of the Wits IDORI Microbe-Cancer Pairs (MCP) team; implemented the team's AI co-scientist pipeline.

Bachelor of Science Honours in Computer Science (Cum Laude) **Jan 2024 - Nov 2024**
University of the Witwatersrand

- Relevant coursework: AI, Machine Learning, Natural Language Processing, Algorithms, Computer Vision, Probabilistic Graphical Models and Robotics
- Research Project: Semi-automated grading of free-text computer science answers

Bachelor of Science in Computer Science (A aggregate) **Jan 2021 - Nov 2023**
University of the Witwatersrand

- Relevant coursework: Mathematics, Computer Science and Applied Mathematics

PUBLICATIONS

- ***Moonsamy, M. Z.***, Dawood, M. M., et al. Democratizing Discovery: Cost-Efficient Scientific Literature Screening with Open-Weight SLMs. SAICSIT 2026, Springer CCIS (to appear). Won *Best Paper Award (CS Track)* at the SAICSIT 2026 conference.
- Kokkas, K., ..., ***Moonsamy, M. Z.***, et al. Artificial intelligence can match domain experts in evidence extraction and critical appraisal of microbial oncogenesis research publications. Frontiers in Cellular and Infection Microbiology, 2026.

WORK EXPERIENCE

Summer@EPFL (École Polytechnique Fédérale de Lausanne) intern, Switzerland **Jun 2025 - Aug 2025**

- Worked under the supervision of Dr Trevor Brokowski at the Laboratory for Intelligent and Humanitarian Global Technologies (LiGHT Lab) on the Mam-AI project.
- Developed the retrieval component of Mam-AI (document processing, chunking and embedding-model experiments), an offline RAG app supporting midwives in Zanzibar during antenatal and postnatal care. Participated in the Kaggle Google Gemma 3n Impact Challenge.

Consultant, GRAI Labs **Apr 2025 - May 2025**

- Worked on a Medical AI project focused on prompt optimization for Vision-Language Models in tuberculosis (TB) classification.

Junior Software Developer, IT Valley Tech **Jun 2023 - Mar 2024**

- Worked as a junior developer to build a full-stack application for EdX using FlutterFlow (low-code platform) for the frontend and Firebase for backend services.

Computer Science Tutor, University of the Witwatersrand **Jan 2023 - Present**

- Tutored courses in NLP, Data Structures, Algorithms, Databases, Machine Learning and Artificial Intelligence, with responsibilities including tutorials, marking, and exam invigilation.

ACHIEVEMENTS

- 3rd Place, Best Research Presentation at Wits Faculty of Science Research Showcase (2026)
- Best Paper Award (Computer Science Track), SAICSIT (2026)
- MSc Fellowship from IDORI (2025 - 2026)
- Postgraduate Merit Award (2023 - 2026)
- Dean's List (2021 and 2024)
- University Council Merit Scholarship (2022)
- Golden Key Honour Society Member (2021)
- University Entrance Scholarship (2021)
- Finalist Wits Mathematics Competition (2021)

TEACHING AND WORKSHOPS

- Gave an online lecture to 80–100 molecular bioscience students at the Wits Sydney Brenner Institute for Molecular Bioscience, explaining how LLMs and transformer models work (2026)
- Facilitator for “Building a Transformer from Scratch” online workshop hosted by Wits MIND Institute and Kempner Institute at Harvard University (2025)

VOLUNTEER WORK

- Volunteered at my former high school as part of community service, supporting the admissions and administration team. This included helping the school transition to online applications during COVID-19, assisting with admissions, and continuing with general administrative duties afterwards, contributing 60 hours over three years (2021-2023)

HOBBIES AND INTERESTS

- Hackathons: Wits HPC (High Performance Computing), Adapt IT, Robocup and more.
- Travelling: I enjoy exploring new places, cultures, and experiences.
- Hiking, running, and fitness: I enjoy staying active through outdoor and physical activities.
- Sketching and drawing: I enjoy creative expression through visual art.
- Sports and gaming: Enthusiastic about both video games and board games.